Measurement Discipline for ML Diagnostics: A Psychometric Framework with a LoRA-Merging Case Study

Anonymous authors
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Abstract

ML diagnostic metrics — scores claimed to predict model behavior, merging outcomes, training dynamics, or capability — are routinely reported as point estimates to three or four decimal places without the measurement-theoretic infrastructure that psychological assessment, at least in its methodological canon, has treated as baseline methodology for decades. We argue that this practice systematically overclaims precision and generality. We introduce a framework that coordinates four measurement commitments — construct articulation, reliability estimation, precision/tolerance, and pre-registered confound decomposition — as joint reporting requirements rather than optional virtues. In the worked example, these commitments are instantiated through ICC, SEM, tolerance schedules, and pre-registered decision rules. We demonstrate the framework through a pre-registered decoder-scale study of spectral LoRA-merging diagnostics ($N = 45$ cross-task adapter pairs on Mistral-7B-v0.3). Under the framework's discipline, the case study does three things: it produces a pre-registered null on per-pair merge prediction under family-confound control; it exposes that the headline statistic's cross-environment numerical tolerance at this sample size is of order 10^{-2}, comparable to its sampling variability, so ordinary third-decimal reporting is not supported under reproducibility testing; and it bounds the diagnostic's reliability with an explicit regime-scope caveat. The case study illustrates the paper's central claim: measurement discipline applied to ML diagnostics changes what can responsibly be claimed — not by decorating point estimates with error bars, but by forcing explicit limits on claims that ordinary reporting would otherwise leave intact.

1 Introduction

1.1 The reporting gap

A recurring pattern in the ML methods literature is to report a diagnostic metric as a point estimate to three or four decimal places without any accompanying description of its measurement properties — without a reliability coefficient that would say how stable the score is across seeds or calibration data, without an explicit tolerance schedule that would say how much of the reported precision is substantive, and without a pre-registered confound decomposition that would distinguish the diagnostic's explanatory share from that of simpler alternatives. Two examples illustrate the pattern; the convergence between them across subfields is the feature we want to draw out.

Consider a representative claim from the LoRA-merging literature. Zhou et al. (2026) report that, across 190 task pairs on ViT-B/32 classifiers, an activation-based pairwise diagnostic predicts merge outcome with $r = 0.572$ (TSV, the strongest individual method among those evaluated). The value is stated to three significant figures. No cross-seed reliability coefficient accompanies it: the stability of the correlation when the activation-calibration set is resampled, or when adapters trained under different seeds are substituted, is not reported. No tolerance schedule states how much of the third digit is substantive and how much is sampling variability or implementation sensitivity. No pre-registered confound decomposition distinguishes the diagnostic's explanatory share from what task-family identity alone might capture; the correlation is presented as if the diagnostic were its sole relevant explanans. The point is not that 0.572 is wrong — it may well be the best available estimate from the reported experiment — but that the reporting convention treats a point-estimate correlation as a self-standing measurement rather than as the central claim of an instrument whose reliability, precision, and confound structure have been independently characterized. We use this example not as a critique of the paper's execution under current norms but as an illustration of the norms themselves: current reporting conventions often permit a predictive correlation to function as the primary evidential object without requiring the accompanying measurement infrastructure that would calibrate its interpretation. The example is useful precisely because it is methodologically serious under current norms: the gap is not bad execution, but the absence of a reporting convention that would require measurement calibration.

A parallel pattern appears in capability evaluation. Benchmark accuracies on standard panels — MMLU, HellaSwag, BIG-Bench-Hard, and their successors — are routinely reported to two or three decimal places of accuracy without cross-seed or cross-prompt reliability coefficients, without explicit tolerance schedules for the reported digits, and without pre-registered confound controls that would distinguish improvements attributable to a claimed intervention from those attributable to prompt-format sensitivity, evaluation-metric choice, or evaluation-set contamination. That the same measurement-theoretic context is missing in a subfield with no methodological contact with LoRA-merging research suggests the gap is not an idiosyncrasy of one community but a general feature of how ML diagnostics are currently reported. Here this second instance functions as a parallel motivating substrate rather than as the empirical case of the present paper, whose worked example is the LoRA-merging diagnostic.

Three specific gaps follow from this pattern. Reliability coefficients in the classical-test-theory sense — ICC, SEM, coefficient alpha or its distribution-based analogues — are nearly absent from ML diagnostic reports. Point estimates are quoted to three or four decimals without uncertainty bounds calibrated to the actual dependence structure of the data. Confound decomposition is rare, and when present is usually conducted post-hoc rather than pre-registered. Each of these gaps on its own is repairable by a careful reviewer; collectively, they produce a literature in which a reader who wants to know how much credence to place in a reported value has no reliable way to calibrate.

1.2 The psychometric analog

Psychological assessment has confronted the same class of problem since at least the 1950s, and has developed a coordinated set of methods to address it. An instrument's reliability is treated as a first-class property, measured empirically and reported alongside any substantive claim (Cronbach, 1951; Shrout & Fleiss, 1979). Standard error of measurement — the standard deviation of measurement error implied by the instrument's reliability — supplies the basis for decimal-place precision claims, calibrated to the instrument rather than quoted by convention. Construct validity — the discipline of defending that an instrument measures what its name claims it measures, distinct from operationalization — operates as a unified theory of scoring inference rather than as an ad-hoc checklist (Messick, 1995; Cronbach & Meehl, 1955). None of these concepts is novel, and none is specific to psychology; what is specific to psychology is the systematic expectation that every reported diagnostic value carries this measurement-theoretic context.

The claim this paper makes is narrower than a comparison of statistical sophistication between fields: ML diagnostic reporting often lacks an integrated measurement argument that ties score meaning, reliability, precision, and confound structure to the substantive interpretation of the reported number. Component methods are present; their coordinated application is not.

ML has adopted some of these tools piecemeal, but rarely as a coordinated reporting discipline. The framework we propose in Section 2 is not a claim of conceptual originality but a proposal for systematic, coordinated application of measurement-theoretic methods to ML diagnostic reporting.

A measurement-discipline literature has emerged contemporaneously across multiple voices. Messing (2026) develops a Total Evaluation Error framework for LLM evaluation pipelines that decomposes pipeline uncertainty into design-choice variance and shrinking-with-N variance, demonstrating on MMLU benchmarking that optimized budget allocation halves estimation error at equivalent cost. The U.S. National Institute of Standards and Technology's voluntary-practices document (National Institute of Standards and Technology, Center for AI Standards and Innovation, 2026) structures benchmark evaluation into define-target, run-evaluation, and analyze-and-report stages; its companion statistical-models document (National Institute of Standards and Technology, 2026) formally endorses generalized linear mixed models for variance decomposition on AI benchmarks, demonstrated on twenty-two frontier LLMs. Bean et al. (2025) survey 445 LLM benchmarks and develop a construct-validity failure taxonomy with eight recommendations and an operational checklist. Camuffo et al. (2026) identify five variance sources in LLM annotation for strategy research and demonstrate twelve to eighty-five percentage-point swings from minor design choices, grounding their protocol in generalizability theory. Romanou et al. (2026) decompose total performance variance under semantics-preserving prompt perturbations and report that perturbations account for up to half of total variance and that 63% of model-pair rankings flip under any single perturbation. Heineman et al. (2025) formalize a signal-to-noise framework over 30 benchmarks and 375 open-weight models, releasing a 900K-result evaluation dataset. McGregor et al. (2025) ranks 26 LLM benchmarks on five risk dimensions under a NIST-aligned risk-management framework. Lior et al. (2025) introduces a method-of-moments resampling recipe for stochastic LLM evaluation, applied to five frontier LLMs.

The shared diagnosis across this corpus is that ML evaluation pipelines carry hidden measurement variance that ordinary reporting does not surface. The adjacent works develop overlapping pieces of a measurement agenda: budget optimization, voluntary reporting practices, construct-validity taxonomies, variance-aware annotation protocols, brittleness diagnostics, benchmark-risk scores, and resampling rules. The present paper's contribution is not the first use of any individual component, but the coordinated application of four measurement commitments to ML diagnostic reporting: construct articulation, reliability estimation, precision/tolerance, and pre-registered confound decomposition. The worked example demonstrates what the coordinated discipline changes in practice: which claims are licensed, which are narrowed, and which must be refused.

1.3 Contribution claims

1. **A coordinated measurement-discipline framework for ML diagnostic reporting** that prescribes reliability-, precision-, and confound-aware reporting as joint requirements rather than optional virtues. The framework's four components — (i) construct articulation, (ii) reliability estimation, (iii) precision and tolerance with explicit per-quantity schedules, and (iv) pre-registered confound decomposition — are not separately novel; the contribution is their integrated application to ML diagnostic metrics, with the discipline that adopting any one without the others silently reverts the report to the direct-readout view of diagnostic scores the framework rejects.

2. **A worked LoRA-merging case study showing how the framework changes scientific interpretation**: a pre-registered per-pair prediction claim becomes an informative null after family-confound control; a task-boundary result is scoped to population-level structure rather than per-pair prediction; a cross-seed reliability estimate is bounded to the same-task regime that generated it; and a pre-registered four-method comparison produces a regime-scope null. The framework's discipline succeeds by making each refusal explicit rather than by rescuing the diagnostic.

3. **A reporting-level precision finding for rank-based residual correlations**: in this small-N residualized setting, partial Spearman ρ exhibits cross-environment numerical drift of order 10^{-2} between equally faithful execution paths, comparable to its sampling variability, so third-decimal reporting is not supported under reproducibility testing. The empirical observation is what measurement-discipline scrutiny surfaces; ordinary diagnostic reporting conventions do not force such findings into view.

In the worked example, the framework changes the scientific interpretation in three specific ways. A diagnostic relationship that an unstructured presentation might overinterpret becomes an informative null once family-pair identity is residualized out (Section 5.1). A reliability estimate that a default report would quote as a universal property of the instrument is shown to license only the regime that generated it (Section 4.2, Appendix D). A headline statistic that default reporting would quote to three decimal places is shown to have effective numerical reproducibility only at its second decimal under ordinary environment variation, turning the third digit into an unsupported claim (Section 6). These are not cosmetic additions to the report; they alter the claims the study is allowed to make.

1.4 Road map

Section 2 develops the framework. Section 3 introduces the worked example's setting. Sections 4–6 present the worked example. Section 7 returns to the framework and generalizes from the case study to broader reporting practice. Section 8 addresses limitations and common objections to the thesis. Section 9 concludes.

The empirical case study is not offered as evidence that spectral diagnostics solve LoRA-merge prediction. It is offered as a stress test of the framework under deliberately unfavorable conditions: a pre-registered diagnostic fails its primary prediction, a strong confound explains most outcome variance, and the headline statistic proves less numerically precise than ordinary reporting conventions would imply. The case is useful precisely because it demonstrates how measurement discipline constrains the claims authors can responsibly make.

2 A Measurement-Discipline Framework for ML Diagnostics

Two views of what an ML diagnostic metric is stand behind the reporting practices this paper addresses. On the first, which operates implicitly in much of the ML methods literature, a score is a direct readout of a latent property: the computation from model weights, activations, or outputs reports what its name claims it reports, and the only remaining measurement question is whether noise has been adequately reduced. On the second, which the psychometric tradition has developed since at least Cronbach & Meehl (1955), a score is a fallible indicator: an observable whose relationship to the theoretical object it is taken to measure must be structurally defended rather than presumed. The framework we propose in this section is what follows from replacing the first view with the second.

What changes here is conceptual, not methodological. The reorientation produces a specific set of reporting practices, but the practices are consequences rather than the starting point. Treating a diagnostic as a fallible indicator commits the author to defending, jointly and inseparably, four things: what the score indicates, how stable the indication is, what precision the indication can actually support, and what else could explain the signal. These are not four independent methodological virtues from among which an author may pick and choose; they are four aspects of a single commitment, continuous with the unified theory of construct validity (Messick, 1989; 1995) that psychometrics has developed in the decades since Cronbach & Meehl (1955). An author who omits any one tacitly reverts to the direct-readout view at the point of omission, since a score whose stability, precision, or alternative explanations go undefended is a score being treated as though those defenses were unnecessary. The framework's internal coherence is the argument; its component practices, taken separately, are the ordinary stock of classical test theory and, individually, could be adopted without challenging the default view of what an ML diagnostic is.

We develop the four defenses in the order a reader's credence should consult them. A reader asks of any score, first what it purports to measure (§2.1), then how stable its readings are (§2.2), then what precision those readings can support (§2.3), and finally what else might be producing them (§2.4). The ordering is epistemic rather than logical — the four defenses are jointly required, and no one of them is prior to the others in the inferential argument — but it tracks the questions a careful reader's credence would need resolved before taking a score as evidence for a substantive claim.

2.1 What the score indicates

A diagnostic's *theoretical object* is what the score is taken to measure: a geometric property of two adapters' updates, a capability's presence or absence, a training regime's character. Its *operationalization* is the specific computation that instantiates the object. Under the inferential view the paper must state both explicitly and defend the connection between them as a separate argumentative step; under the direct-readout view the distinction does not arise, because the operationalization is treated as transparently identical to the object. The typical ML failure mode is the conflation: a paper that says "our spectral alignment score predicts merge outcomes" is usually reporting results about the operationalization (a specific computation from weights) as though they were results about the theoretical object (the geometric relationship between two adapters' learned updates). Under the inferential view the conflation is an error; under the direct-readout view it is not even legible as one.

The distinction has empirical consequences. Different operationalizations of the same theoretical object can produce different predictive patterns on the same data. A subspace-principal-angle-based overlap metric and a singular-value-weighted cosine alignment both claim to measure spectral similarity between two adapters' weight-space updates, but their raw values lie on non-comparable scales, and cross-study comparisons of the raw values are unsound without metric-family harmonization. The psychometric tradition treats such harmonization as part of construct validity in the unified-theory sense (Messick, 1989): the validity argument *is* the defense of the connection between theoretical object and chosen operationalization, and no cross-operationalization claim is licensed without it. Pre-registered articulation of the distinction — stating the theoretical object, the operationalization, and the defense of their connection as three separate things — is the discipline that forces the harmonization to be explicit rather than presumed.

2.2 How stable the indication is

A score has evidential weight only to the extent that it would reproduce under replication of the conditions that produced it. For an ML diagnostic computed from stochastic training processes, the relevant replication is over training seeds, calibration-data samples, and implementation details that vary across labs. An ML diagnostic reported as "$\rho = 0.573$" without an accompanying reliability coefficient is making an implicit claim that this second question has been answered in the limit — that the score is perfectly reliable, in the sense that its reports do not vary across replications — which is almost never true and never licensed by the report itself.

We make the recommendation specific rather than hortatory. For a score computed per-adapter, cross-seed intraclass correlation (ICC; Shrout & Fleiss, 1979) is the natural coefficient, estimated from three or more seeds per task; at two seeds the estimate has confidence intervals whose width is itself informative about the evidential weight of the score's reports. For a score computed per-pair of models, the analog is cross-seed-pair ICC across pairs that should be semantically equivalent (same task-pair, different random seeds). The present study's design commits to three seeds per task specifically to support this estimate. On this design the estimate is $\hat{\rho}_{\mathrm{ICC}} = 0.566$ (moderate) with SEM $= 0.014$ and 95% CI $[0.165, 0.874]$ by Shrout–Fleiss (Appendix D), reported as the instrument-validation step the framework requires.

2.3 What precision the indication supports

A score reported as "-0.533" to three decimals makes an implicit claim that its third decimal carries information. The claim may be defensible from sampling variability alone; it is rarely defensible once implementation sensitivity and cross-environment numerical tolerance of the statistic at the operative sample size are also considered. By *cross-environment numerical tolerance* we mean the observed variation in the reproduced statistic across supported numerical environments, holding the data, analysis specification, and accepted numerical implementation fixed — the empirical scale at which the statistic's reproduction drifts when the same computation is run faithfully under different floating-point paths. Bounded precision is the discipline of stating what the uncertainty on the reported decimal is, calibrated per source. The relevant calibration is not a single "standard error" but a tolerance schedule: sampling variability, implementation sensitivity, and cross-environment numerical tolerance each contribute, and which one dominates depends on the statistic, the operative sample size, and the reproduction condition.

Section 6 develops the paper's most striking instance: the worked example's headline statistic has cross-environment numerical tolerance of approximately ± 0.01 at $n = 45$, meaning its third decimal is not a real claim regardless of what sampling variability the bootstrap confidence interval reports. The observation is not about the bootstrap's calibration; it is about the statistic's cross-environment numerical tolerance on residuals at this sample size. A single "$\pm$" would not have surfaced the distinction. A tolerance schedule does.

2.4 What else could explain the signal

The final defense is the discipline of distinguishing what a diagnostic predicts from what is predictable by simpler alternatives — task identity, input length, surface features, known demographic structure of the data, or in the present paper's worked example, task-family-pair identity. Under the direct-readout view the distinction is not load-bearing because the score's predictive relationship to its object is presumed; under the inferential view the distinction *is* the test of the score's information over and above alternatives. A score whose reported predictive correlation is entirely accounted for by a simpler alternative cannot be interpreted as evidence for its purported object without also specifying that alternative.

Pre-registration of the decomposition is what distinguishes the framework's confound-handling from the post-hoc version that currently dominates. Post-hoc decomposition following a failed main analysis reads as an excuse; pre-registered decomposition is a commitment that the diagnostic must clear *before* seeing the data. The worked example in Section 5 makes the general form concrete with a family-pair residualization and a ΔR^2 threshold; we develop the specifics at the relevant point of the worked example (Sections 4 and 5.1) rather than in advance of it, to keep the framework itself independent of the particulars of any single application.

3 Case Study: Spectral Diagnostics for LoRA Adapter Merging

Low-rank adaptation (LoRA) fine-tuning (Hu et al., 2021) has become a standard method for task-adapting large pretrained models. A practitioner who has accumulated multiple LoRA adapters for different tasks often wishes to combine them into a single adapter that supports all the original capabilities simultaneously. Adapter merging — typically a weighted linear combination of the adapter weight deltas — accomplishes this cheaply but can produce large performance degradation on one or both source tasks.

A recent literature has proposed methods for improving merge outcomes (Stoica et al., 2024; Gargiulo et al., 2025; Li et al., 2026; Zhang & Zhou, 2025) and for predicting which adapter pairs are likely to merge successfully (Zhou et al., 2026; Rahamim et al., 2026; Bolton et al., 2026). Decoder-scale empirical evaluations of these methods have emerged in parallel (Hitit et al., 2026; He et al., 2025), typically reporting predictive correlations as point estimates without the measurement-theoretic context Section 2 requires.

Our worked example is a pre-registered decoder-scale study of one such spectral diagnostic: O-module depth-weighted singular-value-weighted alignment, denoted S_{H1}. The formal definition is in Appendix A. The LoRA-merging setting is not the paper's general target; it is the worked substrate on which the framework is demonstrated. The argument would apply equally to interpretability scores, capability evaluations, or any other ML diagnostic class. This section treats the setting as background just sufficient for a reader outside the merging subfield to follow the worked example.

4 Applying the Framework: Pre-Registration and Design

4.1 Construct articulation for the spectral triage claim

The theoretical object of S_{H1} is a geometric property of two adapters' weight-space updates: specifically, the extent to which their dominant singular directions occupy shared subspaces. The operationalization is a specific computation from per-layer singular value decompositions (Appendix A), weighted by linear depth, restricted to the O-projection module. The choice to focus on the O-projection and to use linear depth weighting derives from precursor analyses (Anonymized, 2026a) which found that O-projection same-cross separation was stronger than Q/K/V and that deeper layers separated same-task from cross-task pairs more sharply than shallower ones. Those observations do not establish that S_{H1} is the best possible operationalization; they license a choice among many plausible ones, and the pre-registration commits to the specific choice before testing.

4.2 Reliability considerations at pre-registration time

The present study's design commits to three seeds per task across eight pre-registered tasks, yielding 24 adapters. This design supports cross-seed reliability estimation for S_{H1} as an instrument: same-task adapter pairs (three same-task pairs per task, 24 total) provide the replicate structure that a cross-seed ICC estimate requires. The resulting estimate, $\hat{\rho}_{\mathrm{ICC}} = 0.566$ with SEM $= 0.014$, is reported at Appendix D; for the purpose of the pre-registered H1 test, the relevant observation is that the design was calibrated to support this estimate before data collection began, not that the estimate was examined post-hoc to reassure reviewers.

The reliability estimate validates only the same-task regime that supplies its replicate structure. The H1 test concerns cross-task pairs, for which the present design contains no direct reliability estimate. The reliability estimate is therefore not a property of S_{H1} in general. It is a property of S_{H1} under the same-task seed-replication regime; the cross-task H1 regime remains reliability-unestimated in this design. We therefore report the same-task ICC as an instrument-validation constraint, not as a universal reliability coefficient for S_{H1}: the estimate licenses the claim that the score is moderately stable when the same task is scored under different seeds; it does not license the claim that the score is moderately stable across tasks. Explicitly naming what the coefficient does not cover is part of the proposed reporting discipline.

4.3 Confound decomposition pre-registration

Four confounds were identified during the precursor analysis and were pre-registered against in the present study's design (Anonymized, 2026c):

- **C1 source-metric dynamic range**: adapters trained to accuracy within a tight band (here, 0.70–0.90) to prevent source-accuracy residual from driving correlations.
- **C2 task-family partition**: eight tasks spanning five task families with no family exceeding two tasks; pre-registered family-pair labels used for residualization.
- **C3 within-task variance**: three seeds per task, enabling reliability estimation and distinguishing within-task variance from cross-task variance.
- **C4 post-hoc fitting**: the H1 score is pre-registered in the spec; no post-hoc search across alternative scores is permitted in the primary test.

The decision rule operationalizes C2 specifically: partial Spearman ρ between S_{H1} and the max-degradation outcome is computed on OLS residuals after both variables are residualized on the pre-registered family-pair labels (FAMILY_B). The ΔR^2 test quantifies the additional variance the geometric score explains beyond family identity. The quantitative shares this decomposition produces — 88.1% of outcome variance captured by family-pair identity alone, +0.003 contributed by S_{H1} beyond it (reported in full at Section 5.1) — are the framework's confound-decomposition discipline made concrete, and are what licenses the subsequent null as informative rather than uninterpretable.

4.4 Decision rule with explicit precision

The pre-registered decision rule requires both: partial $\rho \geq +0.50$ at $p < 0.05$, and $\Delta R^2 \geq 0.10$, with sign constraint absolute (wrong sign fails regardless of magnitude). Both thresholds are calibrated to practically meaningful effect sizes for the triage decision the diagnostic is intended to support, not to statistical-significance thresholds alone. A significant but wrong-signed result is committed *a priori* to be interpreted as a null under the rule, not as a reversed confirmation.

5 Worked Example: Empirical Results

The pre-registered H1 is not confirmed; confirmatory replications pass; the four-method comparison produces a regime null. Pulled together at the level of claims, the framework's discipline licenses some claims about the spectral diagnostic family at this sample size and regime, and refuses others. Table 1 summarizes which claims the joint application of the pre-registered analysis and reproducibility-check evidence supports, and which it forecloses.

Candidate claim	Framework result	Licensed?
S_{H1} predicts per-pair merge degradation	Wrong-signed partial ρ, $\Delta R^2 = +0.003$ beyond family baseline (§5.1)	No
Tested spectral scores detect task-boundary structure	Qualitative same/cross separation replicates across three architectures and two metric families (§5.3)	Yes, at population/task-boundary level only
S_{H1} reliability generalizes across cross-task pairs	ICC estimated only from same-task seed replicates (§4.2, Appendix D)	No; same-task regime only
Partial Spearman ρ reportable to three decimals	Cross-environment numerical drift ≈ 0.012 at $n = 45$ (§6)	No; report approximately -0.53 ± 0.01
Tested static weight-space spectral methods support reliable per-pair merge triage	Four-method comparison: no method's CI excludes random baseline (§5.4)	No, for the tested regime

Table 1: Claims licensed and refused by the framework's joint application to the worked example. Each row pairs a candidate claim with the framework result that bears on it and the resulting licensing status under the pre-registered decision discipline. The table is the case study's prescriptive output: the framework's value lies in distinguishing which claims are licensed, narrowed, or refused under the pre-registered decision discipline.

5.1 H1 outcome under the pre-registered rule

Table 2 reports the pre-registered decision criteria and observed values. The partial correlation is large in magnitude ($\rho = -0.533$, $p = 1.6 \times 10^{-4}$, bootstrap 95% CI under family-pair block resampling: $[-0.825, -0.131]$) but opposite in sign to the pre-registered prediction. ΔR^2 is $+0.003$, far below the 0.10 threshold. Figure 1 shows the underlying scatter with the raw and family-residualized regression lines.

Criterion	Required for H1	Observed	Status
Raw Spearman ρ sign	positive	-0.180	FAIL
Partial ρ (FAMILY_B)	$\geq +0.50$, $p < 0.05$	-0.533 (p=1.6×10^{-4})	FAIL (wrong sign)
ΔR^2 over FAMILY_B baseline	≥ 0.10	$+0.003$	FAIL
FAMILY_B-only R^2	—	0.881	(reference)

Table 2: H1 decision criteria and observed values.

Under the pre-registered rule, H1 is not confirmed: it is a pre-registered null for the claimed triage interpretation, despite a statistically significant wrong-signed association. The result is not a weaker version of the predicted effect; it is the opposite of the predicted effect under the pre-registered sign convention. Treating it as support would require changing the construct interpretation after seeing the data — exactly the kind of post-hoc construct re-articulation the C4 pre-registration discipline forecloses. The wrong-signed partial correlation is hypothesis-generating with respect to future construct re-articulation, but it is excluded from licensing any confirmatory conclusion in this study; follow-up is out of scope for the present paper.

5.2 Why this is an informative null, framework-wise

Task-family-pair identity explains 88.1% of outcome variance on its own; S_{H1} contributes $+0.003$. Without the framework's confound-decomposition discipline, the raw correlation (also negative, smaller magnitude) would have been reported without context, and the reader would have had no way to assess what fraction of any predictive claim was actually family identity operating under a different name. The informativeness of the null is legible because the decomposition is pre-registered; it is opaque otherwise. The $R^2 = 0.881$ is itself a finding — a large effect for the simplest alternative explanation, reportable regardless of how the geometric

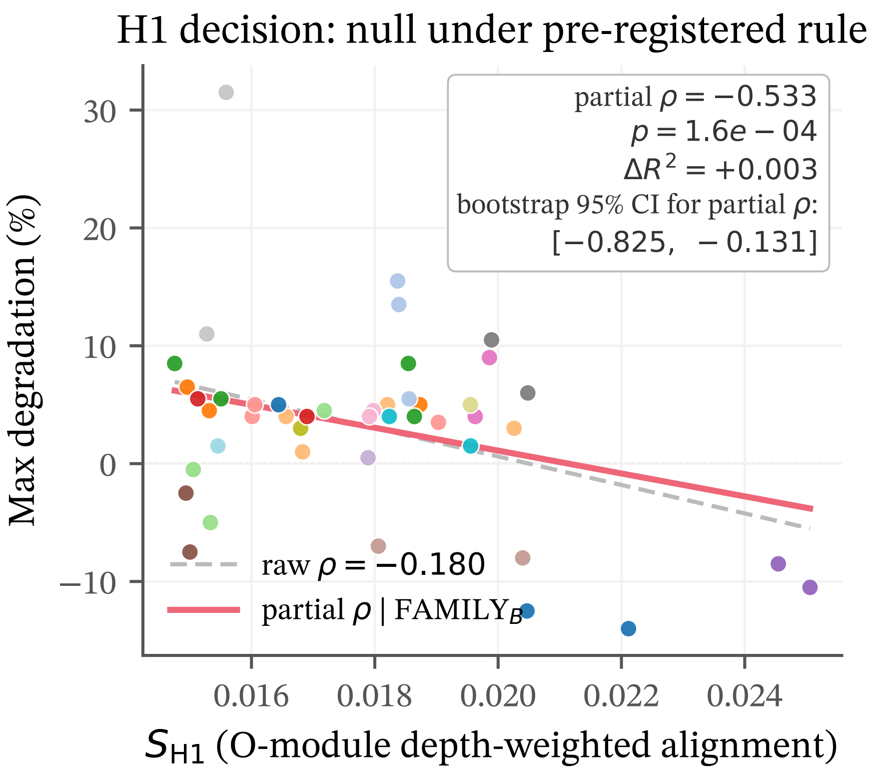

Figure 1: Scatter of S_{H1} against max post-merge degradation for the 45 evaluated cross-task pairs. Grey dashed line: raw OLS regression ($\rho = -0.180$). Red solid line: regression of max degradation on S_{H1} after both variables are OLS-residualized on FAMILY_B dummies (partial $\rho = -0.533$). Point colors encode family-pair identity (28 levels, `tab20` cycled) to make the family-level clustering visible; no individual-level legend is provided because 28 entries would be unreadable at this width. The partial correlation is large in magnitude but opposite in sign to the pre-registered prediction, and therefore fails the sign constraint of the decision rule regardless of magnitude.

diagnostic performs. Because FAMILY_B uses 28 family-pair cells relative to $N = 45$, it is a high-capacity baseline by design: its role in the decomposition is to test whether S_{H1} carries predictive information beyond the pre-registered family-pair structure that the diagnostic would need to exceed in order to support the claimed triage use. The ΔR^2 of $+0.003$ is informative about that specific incremental question; it is not an estimate of what a deployable merge-risk predictor would achieve, and FAMILY_B itself is not offered as one.

5.3 Three-architecture replication of task-boundary detection

Three confirmatory replications of task-boundary detection pass cleanly on the present study (Table 3) and, combined with prior results on DistilBERT (Anonymized, 2026d) and DeBERTa (Anonymized, 2026b), constitute a task-boundary detection replication across three architectures and two metric families.

Claim	Observed	Threshold	Status
B-P1 task-boundary overlap	0/24 same below, 0/252 cross above midpoint	zero overlap	PASS
B-P2 same/cross separation	ratio 2.28×; Welch $t = 15.9$, $p = 6\times10^{-14}$	ratio ≥ 2.0	PASS
B-P4 erank ANOVA across tasks	$F = 165.6$, $p = 1\times10^{-13}$	$p \leq 10^{-6}$	PASS

Table 3: Confirmatory replications, all non-gating for H1 but reported per the pre-registration.

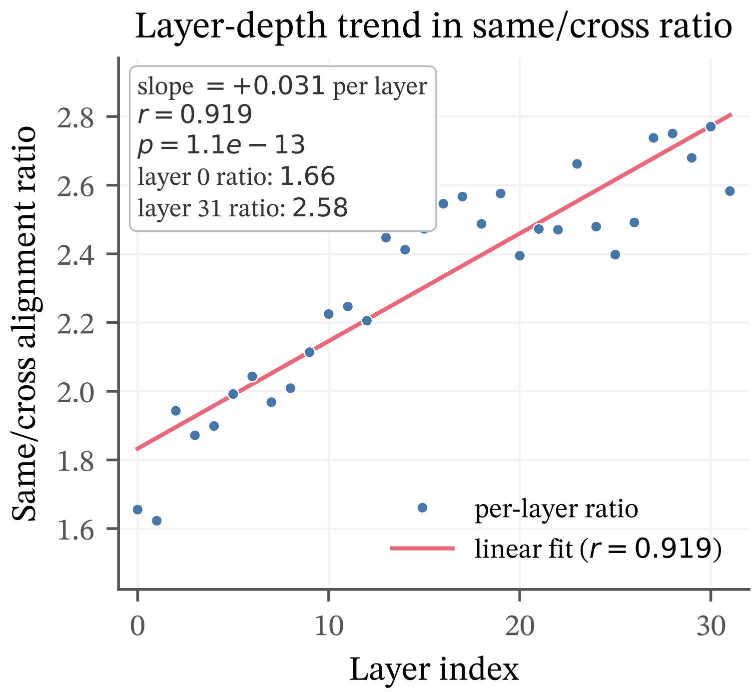

Figure 2: Per-layer same/cross alignment ratio on Mistral-7B in the present study. Linear fit: slope +0.031 per layer, $r = 0.919$, $p = 1.1 \times 10^{-13}$. Ratio rises from 1.66 at layer 0 to 2.58 at layer 31. This is a property of the aggregate population geometry, not a predictor of per-pair merge outcomes: the depth trend motivating S_{H1} (Section 4) survives as an aggregate task-boundary observation about how spectral separation scales with depth, but does not translate into per-pair risk prediction at this sample size and scale (Section 5.1).

The three-architecture claim is framed honestly: two metric families are involved (principal-angle for the precursor studies; SV-weighted cosine for the present one), so same/cross *ratios* are not directly comparable; what replicates is the qualitative structure — same > cross with zero overlap — not any specific numerical scale. The framework's construct-validity discipline is what enforces this restraint. Figure 2 shows the layer-depth trend that motivated the depth weighting in S_{H1}.

Framework-wise, this is a construct-scope refinement rather than a standalone positive result. The positive result is a level-of-analysis result: spectral structure separates task populations, not individual merge-risk pairs in the tested regime. The diagnostic family's construct validity is narrowed by the joint pattern across §5.1 and this section: the spectral score is informative for population-level task-boundary structure, but not for per-pair merge-risk prediction under the tested regime. A report that quoted only the task-boundary replication as evidence for the diagnostic would be operating on a broader construct claim than the evidence warrants; a report that quoted only the per-pair null would miss a real regularity at a different level of aggregation. The framework's discipline is to license each claim at its proper scope.

5.4 Four-method comparison as a regime-scope test

The pre-registered four-method comparison tests whether the regime null is specific to S_{H1} or extends across several static weight-space spectral triage scores. The 45 cross-task pairs are ranked by each of four scores, using pre-registered pairwise-triage adaptations of each method's core quantity rather than the published reference implementations (S_{H1}; KnOTS adaptation (Stoica et al., 2024); TSV adaptation (Gargiulo et al., 2025); SVC adaptation (Li et al., 2026); adaptation details inline in the scripts and in Appendix C). The lowest-risk 22 pairs are retained under each method's ranking and the mean max degradation of the retained set is reported (Figure 3). No method achieves significance at $\alpha = 0.05$; three of four produce wrong-signed

Spearman correlations; KnOTS alone produces a right-signed correlation with positive improvement over random, but its bootstrap confidence interval crosses the random baseline.

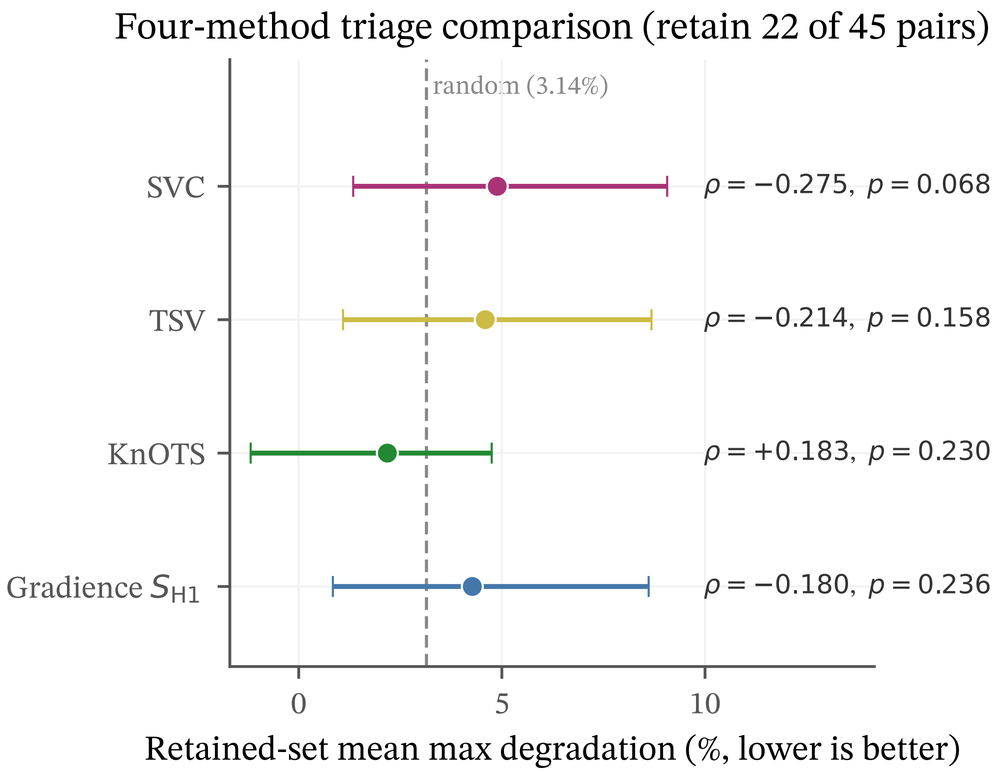

Figure 3: Four-method triage comparison on the 45 evaluated cross-task pairs. Methods are ordered by magnitude of Spearman ρ between their risk score and max degradation. Horizontal error bars are family-pair-blocked bootstrap 95% CIs on the retained-set mean. Vertical dashed line: random-baseline mean max degradation (3.14%) across all 45 pairs. Per-method ρ and p are annotated inline. No method's CI excludes the random baseline. Three of four methods sit visibly to the right of baseline (worse than random on retained-set mean degradation); KnOTS alone sits to the left with a right-signed ρ, but its CI crosses baseline.

The framework reading: within the tested regime — 45 cross-task pairs on a single decoder backbone at a single rank under the specified merge operation — changing the weight-space spectral triage score did not rescue per-pair prediction. The appropriate interpretation is a regime-specific null for this class of static weight-space diagnostics, not evidence against activation-informed, behavioral, or learned mergeability predictors. Activation-informed methods (Zhang & Zhou, 2025; Zhou et al., 2026) remain open.

6 A Reporting-Level Precision Constraint for Rank-Based Residual Correlations

6.1 Observation

Applied to the worked example, the framework prescribes a tiered reproducibility check across a nontrivial environment gap. On execution (T6 in the consolidation record), every scalar in the primary H1 analysis reproduces bit-identical or within float-noise tolerance ($\leq 10^{-3}$) *except* the headline partial Spearman ρ, which drifts from the committed -0.533 to a reproduced -0.545 — an absolute difference of 1.2×10^{-2}. The drift is localized: raw Spearman (no residualization) is bit-identical; R^2 of both the family baseline and the full model is bit-identical; ΔR^2 is bit-identical; bootstrap quantities drift by at most 2×10^{-3}. The only non-trivial drift is in the Spearman statistic applied to OLS residuals.

6.2 Explanation

The likely mechanism is rank instability among near-tied residuals: small floating-point differences in the OLS residuals can change the ordering of observations whose residual values are nearly equal. Because Spearman correlation is computed from ranks, in this observed residual configuration these local ordering changes accumulated into visible second-decimal movement even when raw regression quantities reproduced bit-identically. The exact contribution of any single near-tie depends on the surrounding rank configuration, and we do not attempt a closed-form per-flip accounting; what matters for the paper's argument is that the observed environment-to- environment drift is consistent with this sensitivity. Across numerical environments, the committed value and the reproduced value are *both correct*: each is computed faithfully from the same input data by the same algorithm under different floating-point paths in `numpy.linalg.lstsq`. Neither value is more right than the other. The issue is therefore not computational error, but reporting precision: the statistic is stable enough for the H1 decision and not stable enough to support the displayed third decimal. We therefore treat ± 0.01 as the empirically observed cross-environment numerical tolerance for this statistic in this design, rather than as a universal analytic bound on Spearman correlations on small-N residuals.

6.3 Implication

The H1 decision is robust to this precision: the partial ρ is approximately -0.53 in both environments, decisively below the $+0.50$ threshold (wrong sign) in both, and the decision (H1 not confirmed) is identical in both. But the specific point estimate of partial ρ should be reported as approximately -0.53 ± 0.01 rather than as a four-decimal value. The third decimal is therefore not a supported measurement claim. At larger sample sizes, and especially when residual ranks are less dominated by near-ties, this source of drift should typically diminish; the present study does not estimate a general stabilization threshold.

6.4 Generalization

Rank-based statistics appear widely in ML diagnostic reporting: Spearman correlations, Kendall's tau, rank-based robustness metrics, ranked model comparisons. When these are computed on residuals — i.e., after controlling for another variable via regression — the residualization step can act as a numerical-precision amplifier. Any paper reporting such statistics at small sample sizes without explicitly bounding the precision is making an implicit claim the data does not support. The observation has class-level reach: rank-based statistics on small-N residuals carry a precision risk that this run instantiates rather than uniquely possesses. The present study demonstrates the risk in one controlled setting and motivates reporting it as part of a tolerance schedule rather than assuming that all displayed decimals are substantive.

The observation also demonstrates the framework's central claim: measurement discipline applied to ML diagnostic reporting produces findings that unstructured reporting does not. The precision property was not present in the original analysis and was not present in the committed report; it was surfaced by running the analysis in a different environment and comparing tier-by-tier with a tolerance schedule calibrated to quantity class. Without that discipline, the worked example would have reported -0.533 to three decimals indefinitely, and the third decimal would have been an unstated promise the data could not keep.

Broadening the lens: this instance, together with the two other framework findings the case study has produced — the moderate cross-seed reliability $\hat{\rho}_{\mathrm{ICC}} = 0.566$ (Appendix D) and the confound-decomposition H1 null (Section 5.1) — exhibits a dialectic about what measurement-disciplined reporting yields. The three split into two epistemic kinds. The rank-on-residuals property is a *reporting-relevant measurement constraint*: a fact about how the headline statistic behaves on residuals at this sample size that was not visible in the original analysis and is rarely named in ML diagnostic reports, surfaced here by the framework's reproducibility discipline. The reliability estimate and the informative null are *calibration-and-modesty findings*: the framework's infrastructure does not extend claims but bounds them. Both kinds are framework products, but their epistemic shape differs — one is productive in the sense of generating new claims about measurement, the other in the sense of appropriately bounding old ones. A framework yielding only the first would overreach; one yielding only the second would be merely defensive. The two kinds together are what measurement discipline is *for*.

7 Generalizing from the Case Study

7.1 Reporting practices the framework prescribes

Table 4 summarizes the practical replacement the framework prescribes, organized as a one-to-one mapping between current reporting practice and the measurement-disciplined alternative. None of the replacements is mathematically novel; all are mature methods drawn from classical test theory and its descendants. What the framework adds is coordination — translating the four framework commitments into a concrete reporting bundle rather than applying any one practice in isolation.

Current practice	Measurement-disciplined replacement
Diagnostic claim reported without reliability evidence	Cross-seed ICC or equivalent reliability coefficient reported alongside the claim
Point estimate quoted to convention-set decimal precision	Bounded precision with explicit tolerance schedule, calibrated per quantity class and per sample size (rank-based statistics on residuals require sample-size-calibrated precision bounds)
Raw predictive correlation reported as evidence	Family- or category-residualized partial correlation with a pre-registered ΔR^2 threshold and decision rule
Pass/fail reproducibility check (or none)	Tiered reproducibility verification: qualitative claims, structural agreement, quantitative agreement with tolerance schedule, and gap documentation
Post-hoc analyses reported as confirmatory or decision-supporting	Post-hoc analyses labeled as such and assigned hypothesis-generating rather than confirmatory evidential status

Table 4: Measurement-disciplined replacements for common ML diagnostic reporting practices. Each row replaces a current practice that produces an under-specified measurement claim with a practice that names the measurement design and the evidence the design supports.

7.2 What this changes, and what it does not

The framework does not re-adjudicate specific published claims, but it does argue that readers of the ML methods literature should calibrate credence to the measurement-discipline practices of reports rather than to their reported point estimates. A diagnostic reported as "Spearman $\rho = 0.572$" without a cross-seed distribution, without an appropriate confound decomposition, and without a reproducibility-check trace is making a weaker evidential claim than the reported value alone suggests. This is not a critique of any specific paper. The practices the framework prescribes are not currently normative, so papers not following them are not violating field norms; they are following the norms that exist. Our claim is that the norms should change, and that individual papers should begin applying the framework voluntarily ahead of any field-wide shift. We apply it ourselves in the worked example; we do not ask others to retroactively apply it to work they have already published.

7.3 Why the psychometric analogy matters

The analogy is not that ML diagnostics are psychological tests, but that both are instruments whose observed scores require validity, reliability, precision, and confound arguments before they can support substantive inference. The framework draws from classical test theory and its descendants: Cronbach (1951) on reliability, Shrout & Fleiss (1979) on ICC, Messick (1989; 1995) on construct validity as unified-theory framework, and Cronbach & Meehl (1955) on nomological networks. The contribution is not originality for these concepts but their systematic coordinated application to ML diagnostics: each component is a mature method; their joint use as a reporting discipline is what the present paper proposes.

8 Objections, Alternatives, Limitations

"This is just good statistics; nothing psychometric-specific is needed." Classical test theory contains specific machinery (ICC, SEM, construct validity, convergent and discriminant validity, generalizability-theory decompositions) that is not covered by statistics coursework typical in ML training. The framework is "just good statistics" in the sense that it is not mathematically novel; but it is a specific coordinated application of methods that are not currently coordinated in ML practice. The coordinated application is the contribution.

"Pre-registration is high-overhead and impractical." The worked-example experiment ran end-to-end in ~30 hours for $40 on commercial cloud compute. The pre-registration overhead in this case was modest relative to the experiment: a single committed document before data collection, explicit decision thresholds, and deviation tracking. That overhead is lower than typical reviewer-response cycles in the same literature.

"The case study is weaker than the framework." Two forms: that a null worked example is less compelling than a positive one; that the rank-on-residuals observation is too niche to generalize. To the first: a framework that would have licensed the precursor's false-positive B-P5 claim (Anonymized, 2026a) is worth adopting, and the null demonstrates the framework surviving the outcomes where methodological discipline is most tested. To the second: the rank-on-residuals observation is a demonstration instance, not the paper's center of gravity; the generalization is to rank-based statistics on small-N residuals, which covers a meaningful fraction of diagnostic reporting.

"The cross-environment tolerance result is based on one observed environment gap." Correct. We treat it as an empirical measurement constraint surfaced by the reproducibility protocol, not as a universal bound on rank-based residual statistics. Its role in the paper is to demonstrate why tolerance schedules should be reported, not to establish a general theory of numerical drift.

"The numerical-tolerance observation depends on an environment gap that could have been eliminated by stricter pinning." Stricter pinning would likely reduce this source of variation, and we do not claim otherwise. The reporting issue remains: if a statistic is published as reproducible across supported environments, its displayed precision should be calibrated to the tolerance actually observed under that reproduction condition. In this study, the decision is stable but the third decimal is not.

Limitations proper. The worked example studies a single backbone (Mistral-7B-v0.3), a single LoRA rank ($r = 16$), a single merge operation (0.5/0.5 linear), and a single class of diagnostic metric (weight-space spectral). The framework's claims generalize beyond these specifics; the worked example's empirical claims do not. The direct reliability estimate is also limited to the same-task seed-replication regime; the cross-task H1 regime lacks direct replicate structure and is therefore reliability-unestimated in this design. A reader interested in the framework's application to their own diagnostic class will need to carry out their own application rather than directly importing the worked example's results.

The family-pair residualization used in the H1 test is intentionally conservative and high-capacity relative to the sample size. It is appropriate for testing whether the diagnostic carries incremental information beyond a pre-registered confound structure; it is not appropriate for estimating a general predictive model, and no such model is offered. A lower-capacity baseline (e.g., dummy-coded task-family main effects only, without the full family-pair interaction) would yield a different decomposition and a different ΔR^2; we report the pre-registered specification rather than select among post-hoc alternatives.

9 Conclusion

We have argued that ML diagnostic reporting systematically lacks the measurement-theoretic infrastructure that psychological assessment, at least in its methodological canon, has treated as baseline methodology for decades, that this lack produces overclaims of precision and generality, and that a specific four-component framework — construct articulation, reliability estimation, precision and tolerance with explicit schedules, pre-registered confound decomposition — addresses the gap. The worked example applies the framework to spectral LoRA-merging diagnostics at decoder scale under family-confound control, produces a pre-registered null on per-pair merge prediction, replicates task-boundary detection across three architectures and two metric families, and surfaces an underreported precision limitation of the headline statistic (rank-based correlation on small-N residuals has effective reproducibility of order 10^{-2} across numerical environments) that unstructured reporting conventions do not flag.

Three directions follow. First, activation-informed analogues of the worked example's spectral score remain a candidate for exceeding the regime null the worked example documents. Second, the framework's specific prescriptions will evolve with application experience; this paper is a starting point rather than a completed methodology. Third, individual papers can begin applying the framework voluntarily ahead of any field-wide shift, and we invite them to.

A minimal measurement-disciplined diagnostic report, in the sense this paper argues for, should include: a construct statement that distinguishes the theoretical object from its operationalization; a reliability estimate matched to the regime of intended use; a precision tolerance schedule for the reported quantities; and a pre-registered decomposition against obvious alternatives. A diagnostic that cannot yet supply these elements may still be useful, but its evidential status should be described as exploratory rather than confirmatory or decision-supporting. The purpose of the framework is not to slow ML diagnostics down. It is to make their claims durable enough to matter — so that a reported value, once published, can be read by the field as a bounded measurement rather than as a decimal to be trusted by convention.

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A Exact S_{H1} Score Definition and Decision Rule

For a pair of adapters (a, b) trained on tasks T_a, T_b with $T_a \neq T_b$, let $U_\ell^{(a)}, \sigma_\ell^{(a)}$ and $V_\ell^{(a)}$ denote the left singular vectors, singular values, and right singular vectors of the O-projection of layer ℓ's LoRA update, with analogous quantities for adapter b. Define

$$\alpha_\ell^{(O)}(a,b) \;=\; \frac{\sum_{i,j} \sigma_i^{(a)} \sigma_j^{(b)} \, |\langle u_i^{(a)}, u_j^{(b)} \rangle| \cdot |\langle v_i^{(a)}, v_j^{(b)} \rangle|}{\left(\sum_i (\sigma_i^{(a)})^2\right)^{1/2} \left(\sum_j (\sigma_j^{(b)})^2\right)^{1/2}}.$$

The primary score is the depth-weighted aggregate, $S_{\mathrm{H1}}(a,b) = \sum_\ell \ell \cdot \alpha_\ell^{(O)}(a,b) \,/\, \sum_\ell \ell$ over the 32 layers of Mistral-7B-v0.3.

The decision rule requires both criteria:

1. Partial Spearman $\rho(S_{\mathrm{H1}}^{\perp F}, D^{\perp F}) \geq +0.50$ at $p < 0.05$, where $\cdot^{\perp F}$ denotes OLS residuals after regressing on FAMILY_B dummies, and D is max degradation.

2. $R^2_{\mathrm{full}} - R^2_{\mathrm{family}} \geq 0.10$, where R^2_{family} is the OLS R^2 for $D \sim$ FAMILY_B and R^2_{full} is the OLS R^2 for $D \sim$ FAMILY_B $+ S_{\mathrm{H1}}$.

Sign constraint is absolute: wrong sign fails regardless of magnitude.

B Bootstrap Protocol

All confidence intervals are block-bootstrap resampled with blocks defined by task-family-pair cell (under FAMILY_B). Each resample draws with replacement among the 28 family-pair cells and concatenates all observations within the drawn cells. Reported intervals use 5,000 resamples; percentile intervals at the 2.5% and 97.5% empirical quantiles.

C Pairwise Triage Adaptations of KnOTS, TSV, SVC

The four-method comparison (Section 5) uses pairwise-triage adaptations of each method's core measurement quantity rather than the published reference implementations. These are not implementation-level reproductions of the published systems. They are pre-registered pairwise-triage adaptations of the measurement quantities those papers identify as central to interference or inflation, aggregated at the pair level as a triage score. The faithfulness and departure notes are inline in the score functions' docstrings; full documentation is in the method-comparison script in the supplementary materials. A reviewer who reads "we used KnOTS" in Section 5 should read it as shorthand for "we adapted KnOTS's shared-subspace interference-norm quantity to the pairwise triage setting," not as a claim that we executed the published paper's full pipeline.

D Cross-Seed Reliability for S_{H1}

Cross-seed ICC(2,1) for S_{H1}, estimated from the 24 same-task adapter pairs (three within-task seed-pairs $\times$ eight tasks; absolute agreement, single measurement (Shrout & Fleiss, 1979)), is $\hat{\rho}_{\mathrm{ICC}} = 0.566$, with 95% CI $[0.165, 0.874]$ by the Shrout–Fleiss F-distribution method. A secondary block-bootstrap over tasks (5,000 resamples, seed 134) yields CI $[0.141, 0.965]$, which diverges from the parametric CI by up to 0.091 on the upper bound. We report both intervals. The divergence is itself informative about parametric-assumption strain at $N = 8$ tasks and illustrates that a point estimate with a single CI can understate the uncertainty about a reliability coefficient at small sample sizes; this is the kind of limitation the paper argues should be stated rather than hidden.

The standard error of measurement, $\mathrm{SEM} = \mathrm{SD}_{\mathrm{pooled}}\sqrt{1 - \hat{\rho}_{\mathrm{ICC}}} = 0.014$, is the measurement-theoretic precision of a single-seed-pair same-task S_{H1} estimate. Against pooled same-task S_{H1} values ($\mathrm{SD}_{\mathrm{pooled}} = 0.021$, range $[0.044, 0.112]$, width ≈ 0.068), SEM of 0.014 is roughly 20% of the same-task signal width: substantial, and consistent with the conventional-descriptor reading that reliability here is moderate rather than high. It supports modest precision claims in the same-task regime.

The cross-task regime is where the primary H1 test lives, and it is a different object. Cross-task S_{H1} values occupy $[0.015, 0.025]$, width ≈ 0.010 — an order of magnitude narrower than the same-task range, and *narrower than the same-task SEM itself.* A naive transfer of the same-task SEM to cross-task pairs would yield a self-defeating statement: the instrument could not distinguish any cross-task pair from any other at this precision. That conclusion is not a valid inference from the present design; the primary H1 test nevertheless recovers a significant wrong-signed partial association on cross-task pairs. The appropriate conclusion is that the transfer fails, not that the cross-task regime has been validated: the reliability we have estimated is a property of the instrument in the same-task regime, not a universal property.

What this reliability design can license and what it cannot is therefore worth stating directly. The present design supplies cross-seed replicate structure only on same-task pairs (three seeds per task, yielding three same-task seed-pairs per task). The 45 cross-task pairs used in the primary H1 test are each a single seed-pair draw; the design contains no replicate structure within cross-task cells. No direct reliability estimate for cross-task S_{H1} is therefore reportable within this study. Whether the cross-task regime has comparable or materially different reliability properties relative to the same-task regime is genuinely open, and any future study that wants a cross-task reliability estimate would need a design extension: either more seed-pair samples per cross-task cell (which scales the adapter budget multiplicatively), or an analytical argument that cross-task reliability is bounded above or below by the same-task estimate (which we do not attempt here).

Explicitly bounding the coefficient's regime of validity is part of the proposed reporting discipline (§2.2). The temptation to extrapolate a reliability coefficient beyond the design that produced it is precisely the kind of measurement-theoretic over-reach the paper argues against; refusing the extrapolation in the appendix where the estimate is reported, rather than in a reviewer-response document or a post-publication correction, is the appropriate location for that refusal.

$\hat{\rho}_{\text{ICC}} = 0.566$ places the instrument at "moderate" cross-seed reliability by the conventional descriptive thresholds (> 0.9 excellent, > 0.75 good, > 0.50 moderate, < 0.50 poor). We report the conventional description only as orientation and not as a binary gate; the paper's framework argues against thresholding continuous reliability estimates into verdicts. Full panel, mean-squares decomposition, and bootstrap details are in the supplementary results bundle; design commitments are documented in the pre-registration spec included with the supplementary materials.

E Full Reproducibility-Check Trace

Summary of the reproducibility check performed on the worked example's submitted artifact bundle, organized by the four-tier verification protocol specified in Section 6.

Tier 1 (qualitative claims). All six qualitative claims reproduce: H1 not confirmed; B-P1, B-P2, B-P4 replications all confirmed; no precursor-study composite score exceeds H1 thresholds on the present data; composite-score sign pattern unchanged across the environment gap.

Tier 2 (structural agreement). `analysis_h1.json` and `analysis_secondary.json` schemas reproduce exactly.

Tier 3 (quantitative agreement). All scalars within the amended tolerance schedule. One value (partial Spearman ρ) falls into the rank-on-residuals precision regime, drifting by 1.18×10^{-2} between the original execution environment (-0.5330) and the verification environment (-0.5448). Every other scalar is bit-identical or at float-precision-noise magnitude. The four-method comparison is not fully re-derived from the lightweight submitted bundle because auxiliary per-adapter SVD factor files were excluded for size reasons (1.2 GB). The bundle therefore includes the derived `method_comparison.json` as the canonical artifact for that comparison, and the reproducibility trace records this limitation explicitly.

Tier 4 (gap documentation). Environment gap: Python 3.11 $\rightarrow$ 3.14; `numpy` 1.26 $\rightarrow$ 2.4; `scipy` 1.13 $\rightarrow$ 1.17. Data-availability gap: the auxiliary per-adapter factor files used for the four-method comparison were not retained in the submitted bundle. Neither gap affects qualitative claims.

F Deviations from Pre-Registration

Six code-level or infrastructure-level deviations were caught before H1 was computable from committed data. None altered the pre-registered protocol, task set, training design, S_{H1} definition, or decision rule. Summary: (D1) transformers API change; (D2) pair-key ordering mismatch between `pair_sample` and analysis script; (D3) layer-indexing and module-string schema mismatches; (D4) JSON bool serialization; (D5) CPU-contention incident during pilot training, with loss-trajectory continuity verified post-incident; (D6) per-pair audit file shape gap, resolved via a lazy per-pair synthesizer before any method-comparison numbers were committed. The above D1–D6 summaries are the load-bearing deviation record for review purposes.

G Compute, Environment, Artifacts

Single RTX 6000 Ada 48GB on commercial cloud. End-to-end compute approximately 30 hours at approximately \$40 total cost. Execution environment: Python 3.11, `torch` 2.3, `transformers` 4.44, `peft` 0.12, `numpy` 1.26, `scipy` 1.13. A complete package freeze was not captured before the environment was decommissioned. The supplementary materials accompanying this submission include the pre-registration documents, the four top-level analytical JSONs (H1, ICC, secondary, four-method comparison), the twenty-four per-adapter raw audits with their meta-JSONs, and the three analysis scripts that produce the headline numbers.